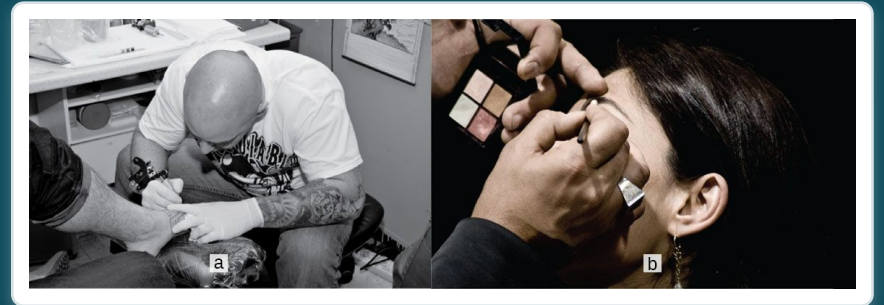


CHAPTER 5

The Integumentary System

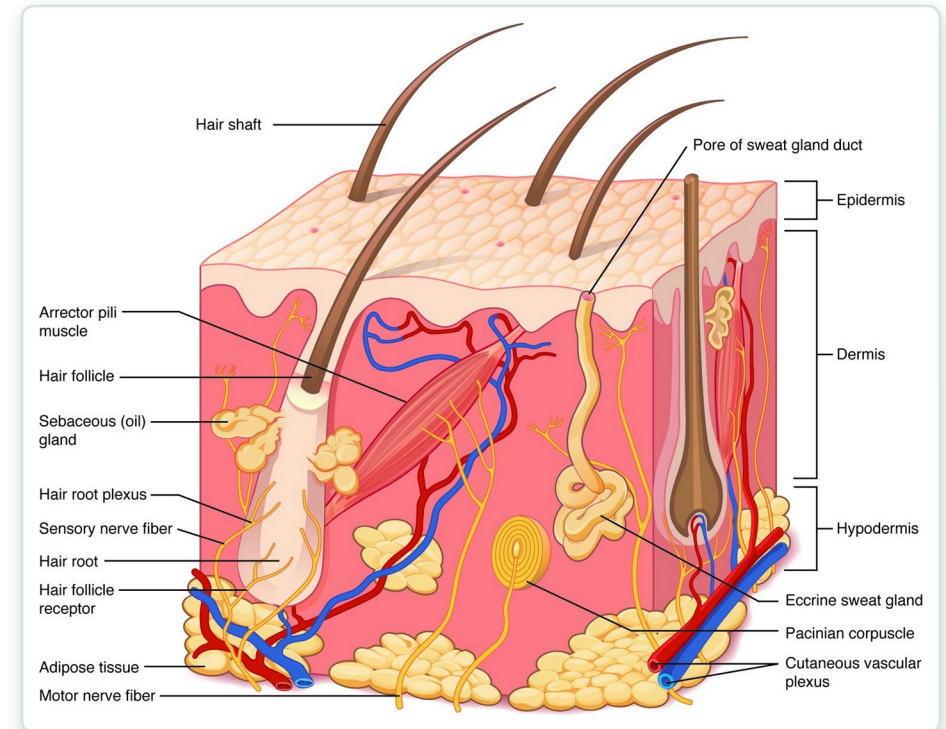
The skin, its accessory structures, functions, and disorders.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the integumentary system and the role it plays in homeostasis
- 2 Describe the layers of the skin and the functions of each layer
- 3 Describe the accessory structures of the skin and the functions of each
- 4 Describe the changes that occur in the integumentary system during the aging process
- 5 Discuss several common diseases, disorders, and injuries that affect the integumentary system



The skin — the body's largest organ.



5.1

SECTION

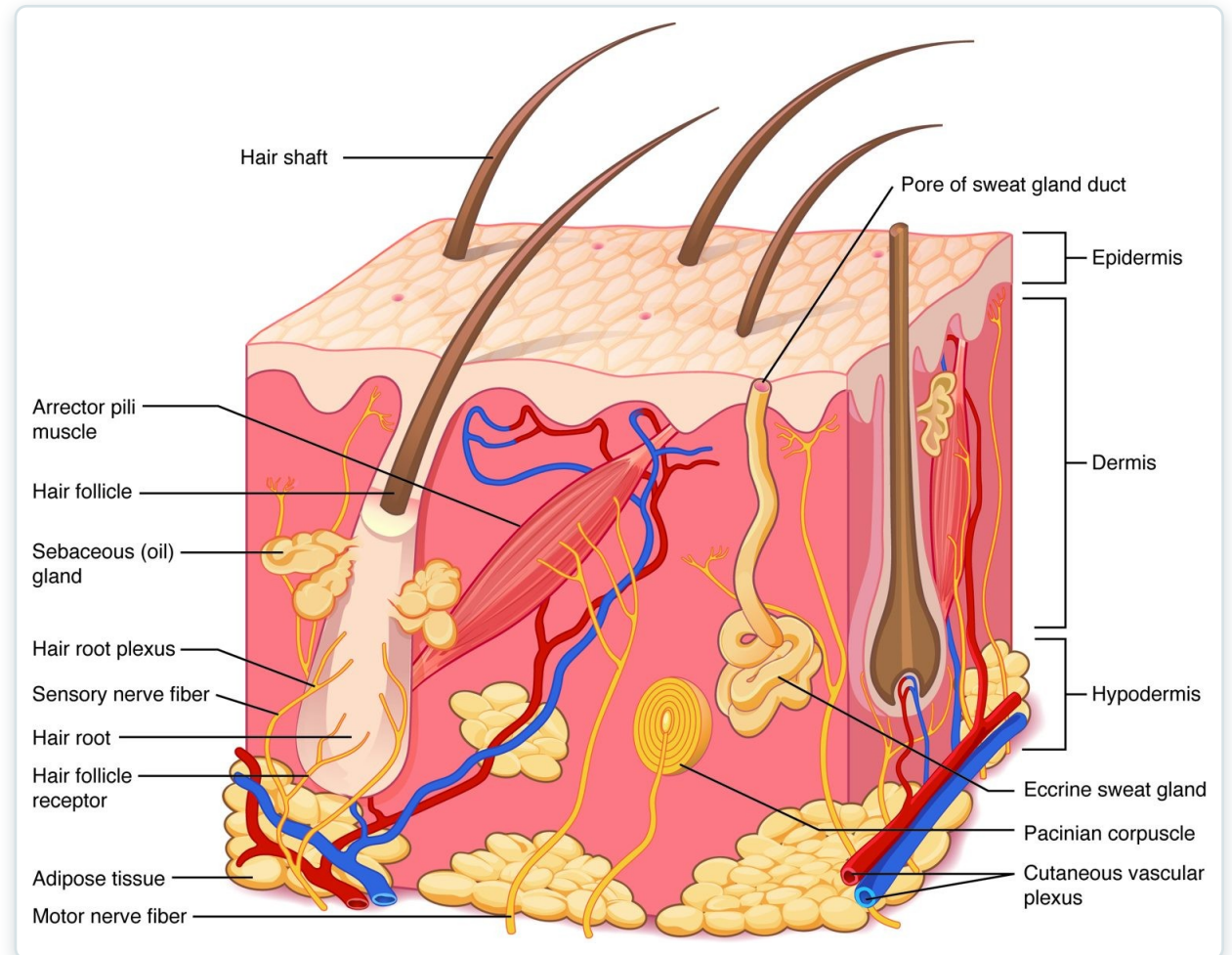
Layers of the Skin

The epidermis, dermis, and hypodermis — the layered structure of the skin.

SECTION 5.1

Structure of the Skin

- Three layers: epidermis, dermis, hypodermis
- Epidermis is avascular epithelium
- Dermis holds vessels, nerves, glands
- Hypodermis stores fat and anchors skin

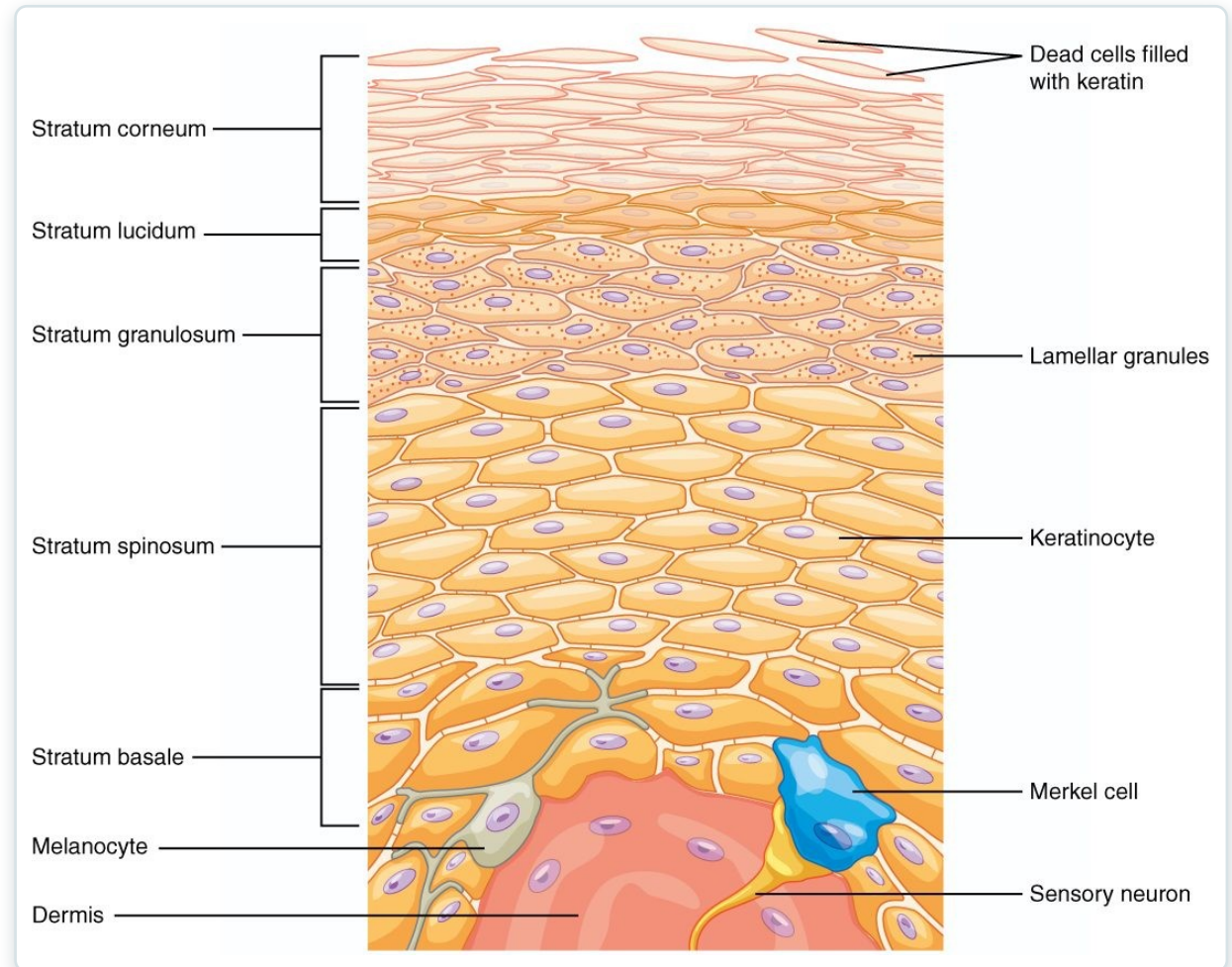


The layers of the skin and underlying tissue

SECTION 5.1

Layers of the Epidermis

- Up to five layers in thick skin
- Stratum basale up to stratum corneum
- Keratinocytes fill with keratin and shed
- Melanocytes and immune cells reside here

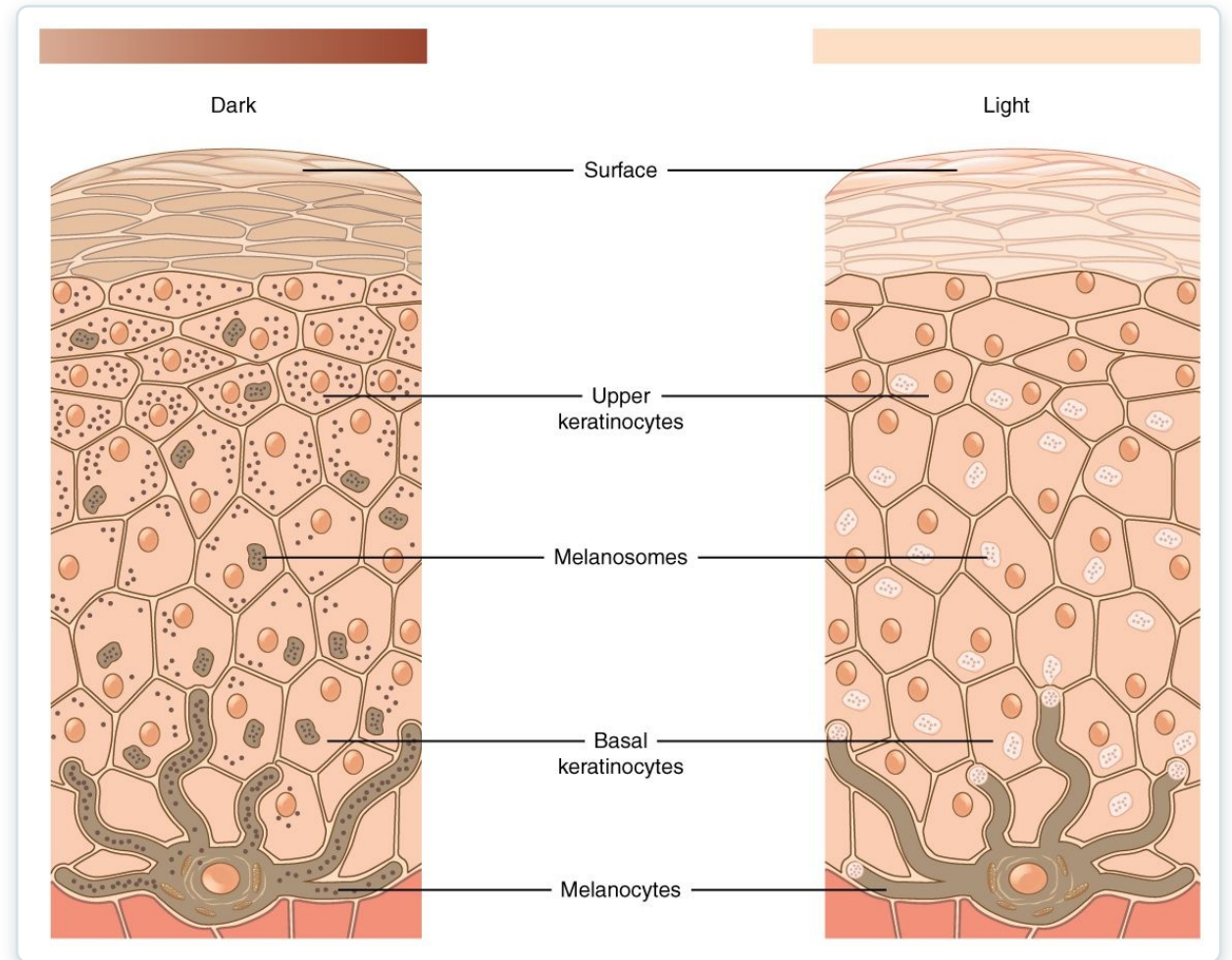


The strata of the epidermis

SECTION 5.1

Skin Pigmentation

- Melanin determines skin color
- Made by melanocytes in the basale
- Protects DNA from UV damage
- Everyone has similar melanocyte numbers



Melanin production and skin color



5.2

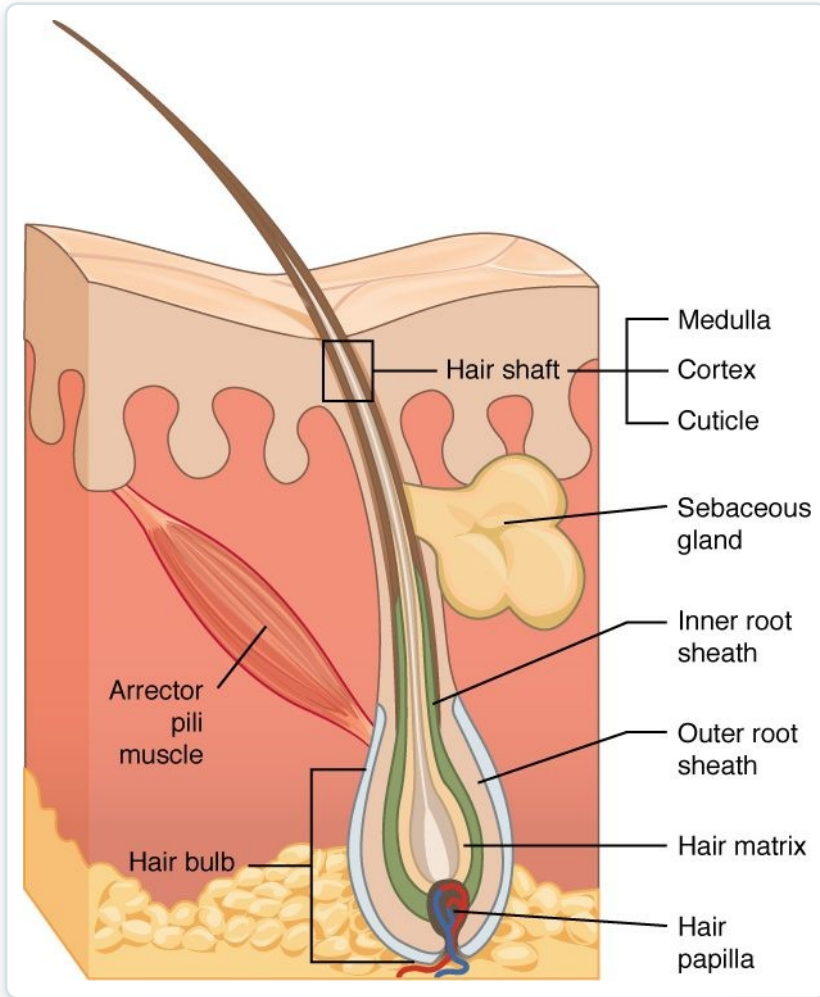
SECTION

Accessory Structures of the Skin

Hair, nails, and glands — the structures that arise from the skin.

SECTION 5.2

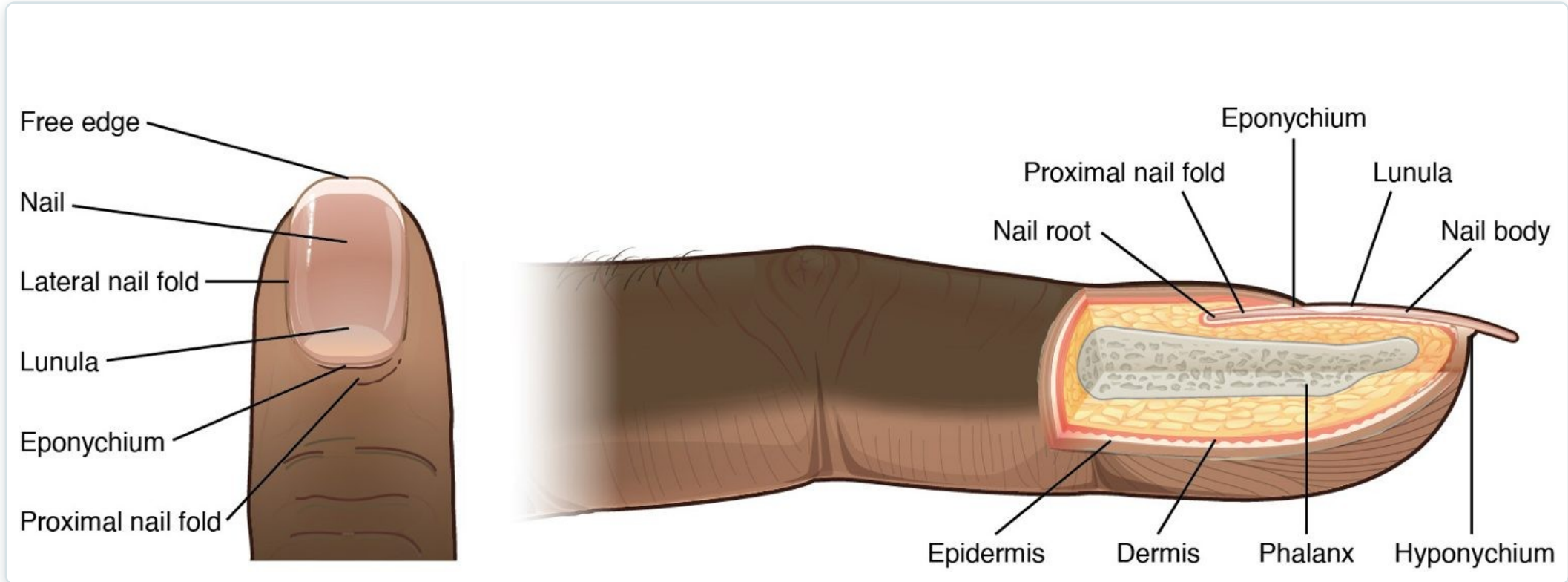
Hair and the Hair Follicle



Structure of a hair and its follicle

- Hair grows from follicles in the dermis
- Shaft is visible; root is embedded
- Arrector pili muscle causes goosebumps
- Provides protection and sensation

Nails



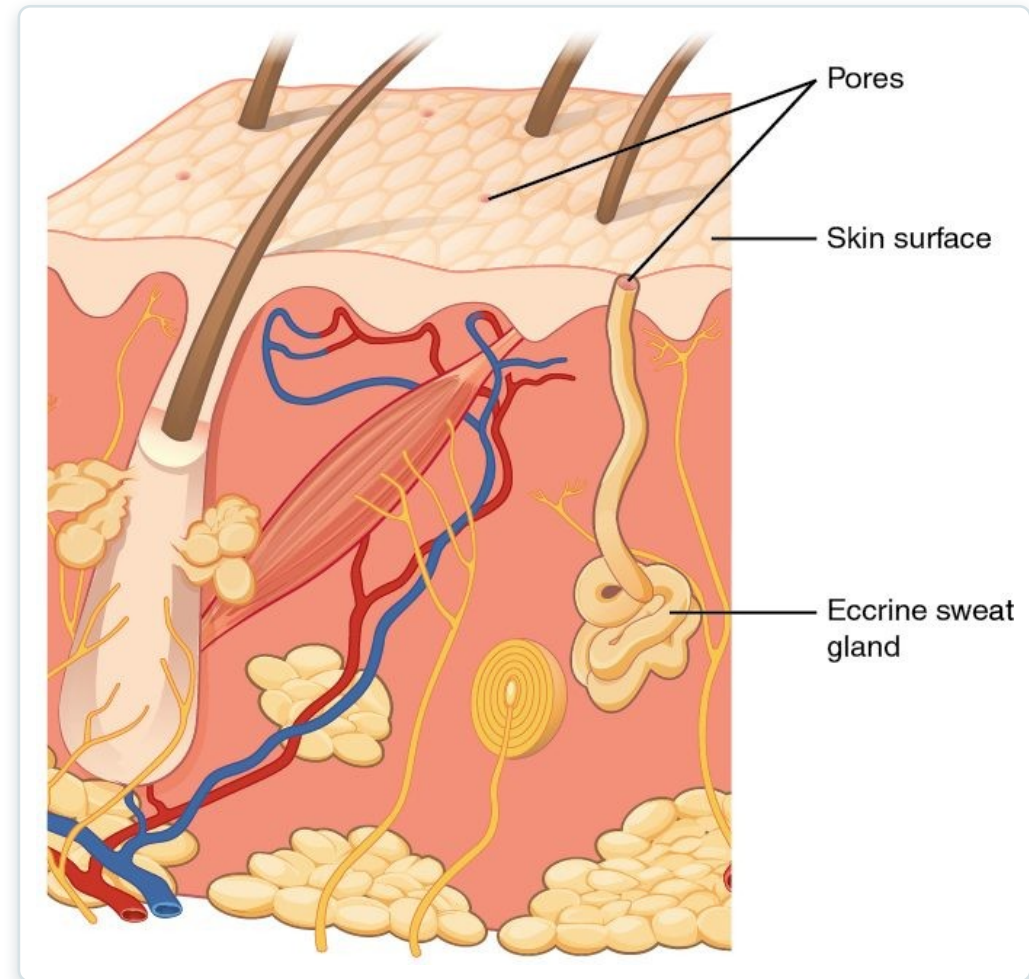
Structure of a nail

- Keratinized plates over the digits
- Protect fingertips and toes
- Nail bed, lunula, and cuticle
- Useful in clinical assessment

SECTION 5.2

Glands of the Skin

- Eccrine sweat glands cool the body
- Sweat evaporates to lose heat
- Sebaceous glands secrete oily sebum
- Sebum lubricates skin and hair



Sweat and sebaceous glands of the skin



5.3

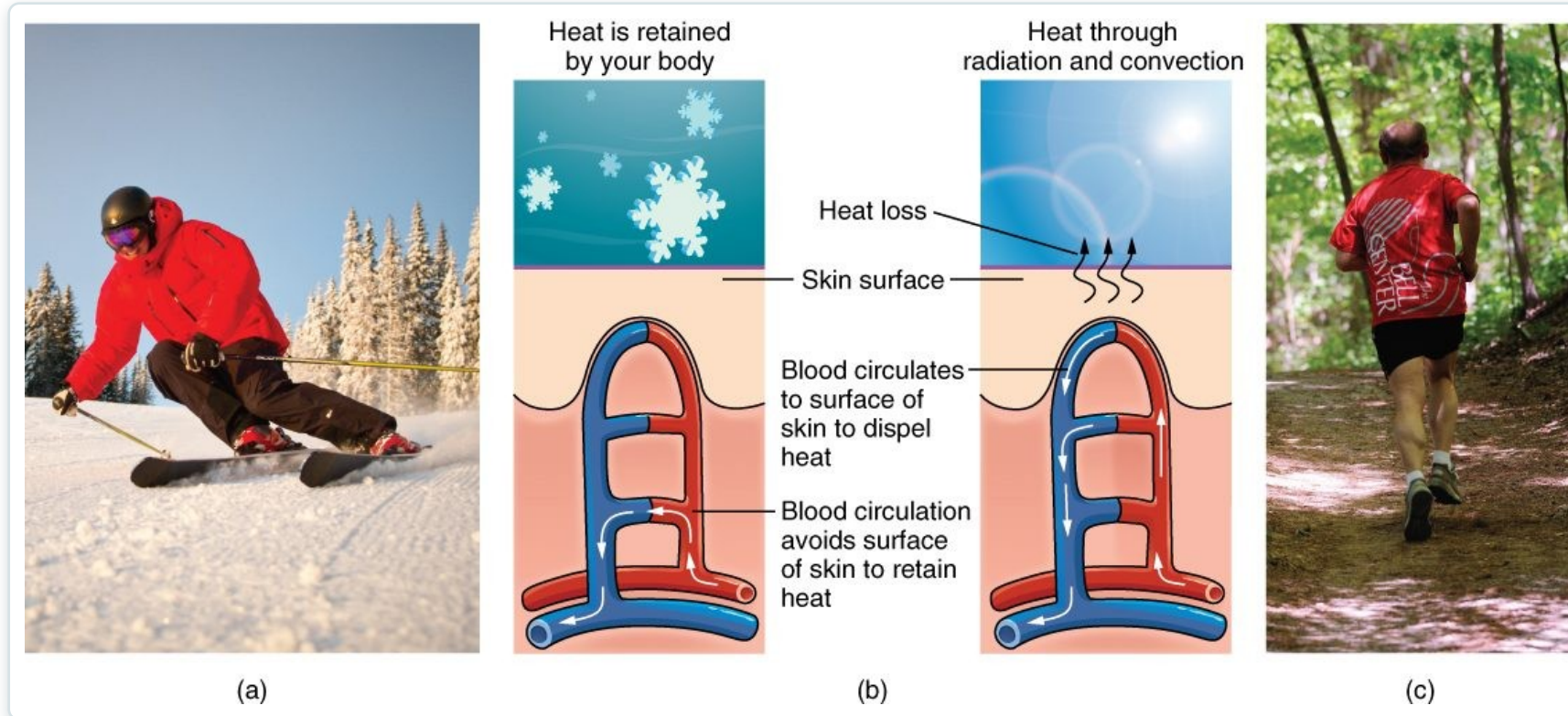
SECTION

Functions of the Integumentary System

Protection, sensation, thermoregulation, and vitamin D — what the skin does.

SECTION 5.3

Thermoregulation



The skin's role in thermoregulation

- The skin helps control body temperature
- Sweating and evaporation cool the body
- Vasodilation releases heat
- Vasoconstriction conserves heat



5.4

SECTION

Diseases, Disorders, and Injuries of the Integumentary System

Burns, skin cancer, acne, and other disorders of the skin.

SECTION 5.4

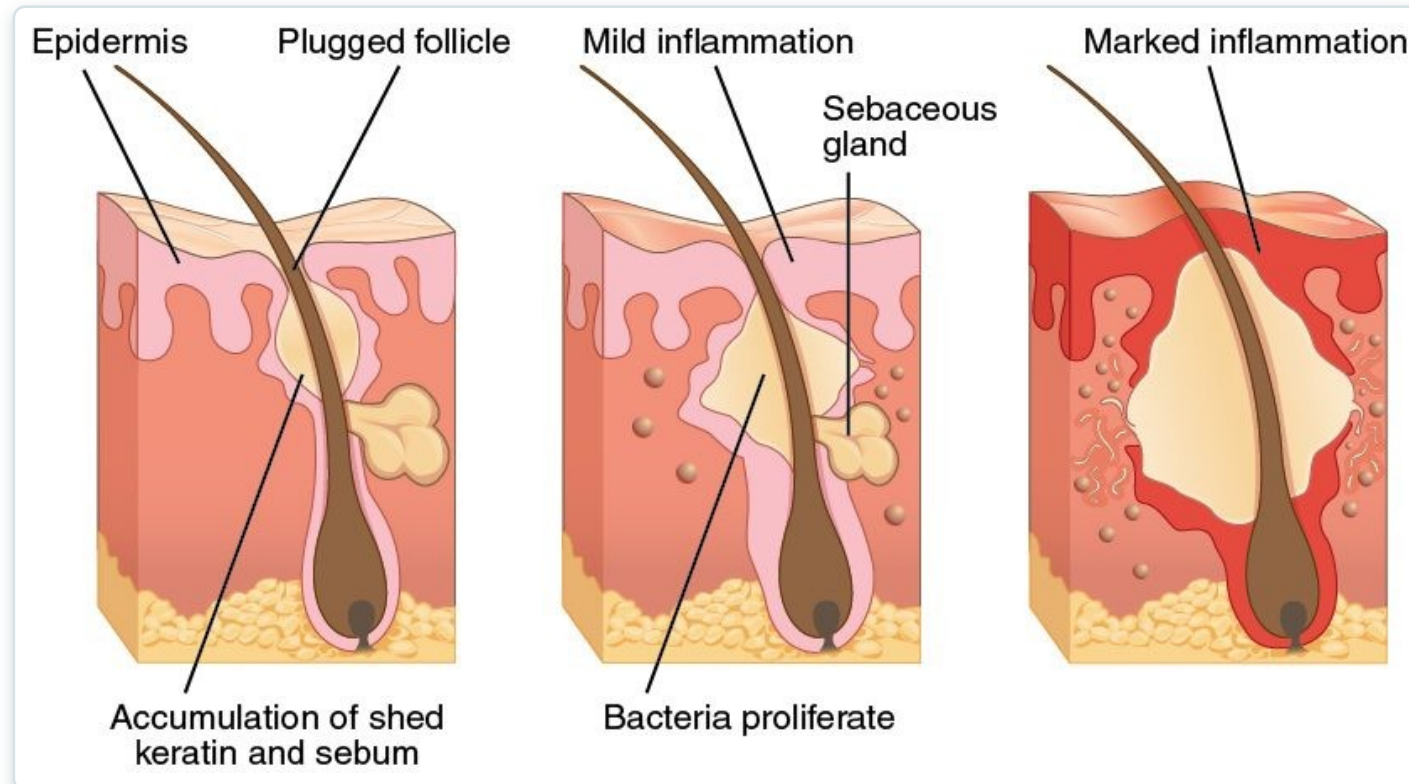
Skin Cancer & the ABCDE Rule

- Three main types of skin cancer
- Basal cell, squamous cell, melanoma
- Melanoma is the most dangerous
- Watch moles for ABCDE changes



Moles and the ABCDE signs of melanoma

Acne

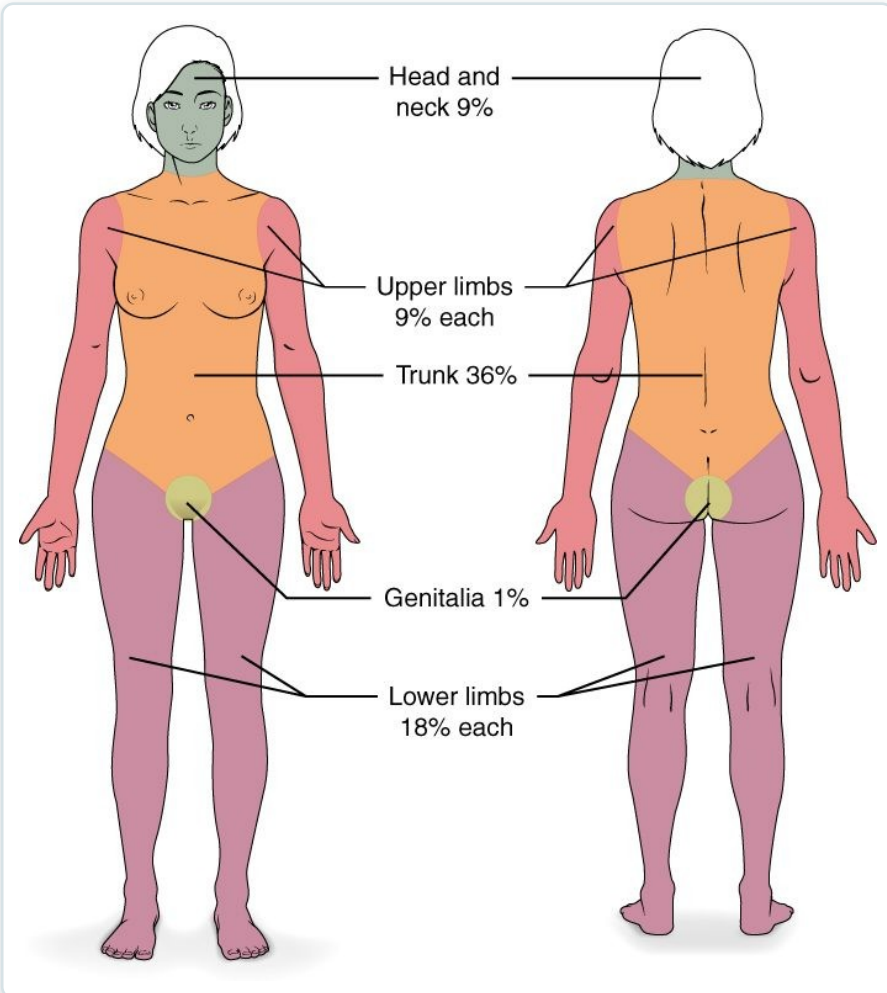


The development of acne

- A follicle becomes plugged with sebum
- Bacteria multiply and inflame it
- Ranges from comedones to cysts
- Common during hormonal changes

SECTION 5.4

Burns and the Rule of Nines



The rule of nines for estimating burn area

- Classified by depth and extent
- Superficial, partial, and full thickness
- Rule of nines estimates body area
- Guides fluid resuscitation

Key Takeaways

- The skin has three layers — epidermis, dermis, and hypodermis — and is the body's largest organ.
- Keratinocytes and melanin give the epidermis its protection and color.
- Hair, nails, and glands are accessory structures that arise from the skin.
- The skin protects, senses, makes vitamin D, and regulates temperature.
- Burns, skin cancers, and infections are common, clinically important disorders.

CLINICAL CONNECTIONS

Why the Skin Matters in Practical Nursing

Skin Assessment

Color, temperature, turgor, and lesions reveal a patient's overall status.

Pressure Injuries

Prolonged pressure breaks down skin — prevention is vital in immobile patients.

Burn Care

Depth and the rule of nines guide fluid resuscitation and treatment.

Infection Barrier

Intact skin is the body's first defense against pathogens.

Skin Cancer

Teach the ABCDE rule and the importance of sun protection.

Thermoregulation

Sweating and skin blood flow help control body temperature.